

A Graph-Based Control Interface for Traffic Signals on Heterogeneous Road Networks

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Abstract

Traffic-signal phases differ with junction geometry, lane connectivity, and signal design, which makes a fixed neural action head difficult to reuse across road networks. This report describes a movement-centric interface that separates representation, learned prioritization, and local action construction. Directed LANE_{GROUP} nodes hold corridor state and MOVEMENT nodes represent legal controlled turns. A shared typed message-passing network produces one scalar score per MOVEMENT. At each junction, a fixed incidence matrix aggregates those scores into logits for phases constructed automatically from SUMO controlled links and conflict data. The learned parameter shapes therefore do not depend on network size or local phase count. PPO experiments support the implementation: policies transfer across synthetic grid sizes and aspect ratios, execute on five heterogeneous city scenarios, and expose a clear limitation when signal coverage changes outside the training distribution. The result is a practical representation and control pipeline, not a claim of a new reinforcement-learning algorithm or universal traffic-control generalization.

1 Motivation

A traffic signal selects a local combination of compatible traffic movements. The number and meaning of those combinations vary: a three-arm junction, a regular four-arm junction, and an intersection with dedicated turn lanes need not share an action schema. Consequently, phase index 2 at one controller has no intrinsic relationship to phase index 2 at another. A policy whose output weights correspond directly to fixed phase indices embeds assumptions about one predefined junction layout.

This implementation instead uses a MOVEMENT—a legal controlled turn from an incoming corridor to an outgoing corridor—as the interface between shared learning and local signal operation. Movements retain common semantics when their number and compatible combinations change. The learned model can therefore emit a variable number of MOVEMENT scores, after which each junction maps its own scores to its own phase set.

The report contributes an integrated implementation:

1. a city-level graph of directed LANE_{GROUP} and MOVEMENT nodes;
2. typed communication for upstream demand, downstream supply, and unsignalized pass-through context;
3. a shared two-block message-passing policy with a scalar MOVEMENT head;
4. automatic construction of junction-specific, conflict-derived phase sets; and
5. an OSM-to-SUMO, PPO-training, and evaluation workflow.

Movement pressure and movement-structured control have substantial prior lineage. Max-pressure methods score compatible stages from constituent traffic flows [1]; PressLight connects pressure to learned control [2]; and FRAP structures a policy around movements and phase competition [3]. TransferLight also maps movement representations to variable phase spaces, using a learned hierarchical phase encoder [4]. The present report does not claim to originate movement scoring. Its focus is the transparent combination of a city-level typed graph, a scalar learned interface, and phases generated by the network-construction pipeline.

2 Representation and control interface

2.1 From road corridors to a typed graph

A LANE_{GROUP} is a directed corridor between controller-relevant endpoints. Opposite travel directions are distinct nodes because they have different queues, speeds, capacity, and destinations. Unambiguous continuations through unsignalized junctions may be contracted; contraction stops before a branch would create false reachability. Dynamic features summarize the detector region near the downstream end, including vehicle and queue counts, occupancy, speed, storage, arrival and departure rates, and queue-tail arrival estimates. Static features describe corridor scale.

A MOVEMENT joins one input LANE_{GROUP} to one output LANE_{GROUP} through a controlled junction. It holds turn-local features such as demand, turn class, controlled-link count, and recent green state. Junctions own Movements and phases but are not graph nodes.

Every MOVEMENT has four directed relation types:

$$L_{\text{in}} \rightarrow M, \quad L_{\text{out}} \rightarrow M, \quad M \rightarrow L_{\text{in}}, \quad M \rightarrow L_{\text{out}}.$$

Message direction is information flow rather than vehicle flow. In particular, $L_{\text{out}} \rightarrow M$ returns downstream storage and spillback context to a turn that would feed that corridor. At an unsignalized junction, legal pass-through

connections become weighted LANEGROUP-to-LANEGROUP edges instead of controllable MOVEMENT nodes. Their weight decays with connector free-flow time t_{ff} :

$$w = \exp(-t_{\text{ff}}/30 \text{ s}).$$

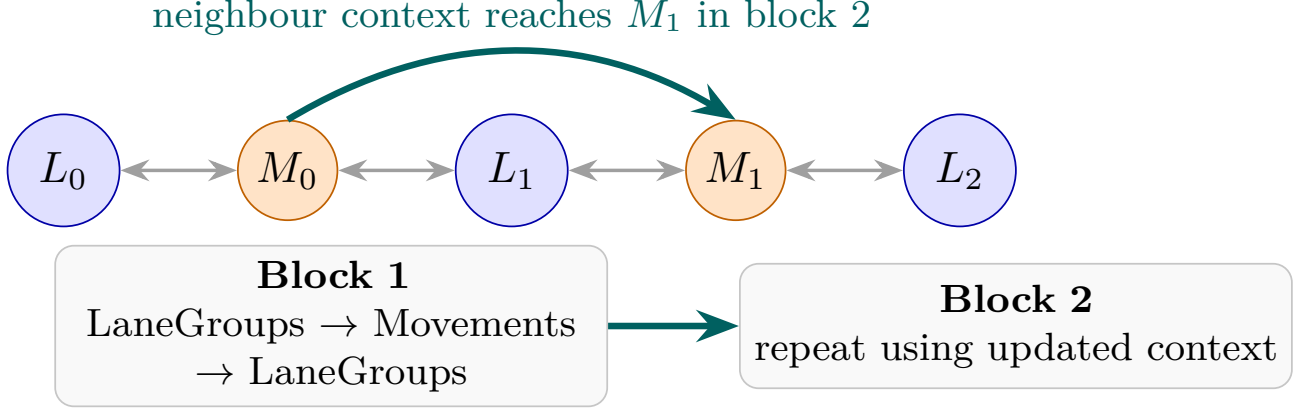


Figure 1: The typed bipartite representation and why the second update block matters. The first block updates Movements from endpoint LaneGroups and returns them to LaneGroups. In the second block, a Movement can use context deposited by another Movement through a shared corridor.

2.2 Two update blocks, not graph attention

The active model is a typed mean-aggregation graph neural network, not a graph attention network. LaneGroups and Movements are embedded separately. For update block k , the essential computation is

$$M^{k+1} = \phi_M(M^k, \text{mean}_{\text{in}}(L^k), \text{mean}_{\text{out}}(L^k)), \quad (1)$$

$$L^{k+1} = \phi_L(L^k, \text{mean}_{M \rightarrow \text{in}}(M^{k+1}) + \text{mean}_{\text{conn}}(L^k), \text{mean}_{M \rightarrow \text{out}}(M^{k+1})). \quad (2)$$

Each relation has its own linear transformation; neighbours are averaged rather than assigned learned attention weights.

Two blocks are used. The distinction is important: initialization and the first block give a Movement its endpoint corridor state, while the second block is the first opportunity for one Movement to affect another through a shared LANEGROUP (Figure 1). The depth is configurable, but this report neither assumes that more blocks are always beneficial nor presents an optimal-depth claim. The final Movement embeddings are mapped to scalar scores s_m . The critic mean-pools the Movements owned by a junction and estimates one value per controller.

2.3 Automatic phases and local action mapping

OSM tags do not directly define the action space used by the policy. The build pipeline uses SUMO `netconvert` [5] to obtain lane connections, controlled-link indices, request indices, and request-conflict (`foes`) data. Links that must activate together are first combined into atomic groups. An atomic group is rejected if it contains an internal conflict. Two groups are incompatible when SUMO reports a conflict in either direction or when different incoming approaches merge into the same outgoing edge.

The compatible groups form a graph. Bron–Kerbosch enumeration [6] returns every maximal clique, and each maximal compatible set becomes one selectable green phase. “Maximal” matters: a smaller protected-turn set remains an action when no additional compatible group can be added, even if another phase contains more Movements. This is conflict-derived construction under the implemented SUMO and merge rules, not formal verification of a real signal plan.

For junction j , let $A_j \in \{0, 1\}^{|P_j| \times |M_j|}$ record which local Movements are green in each selectable phase. The shared GNN produces $\mathbf{s}_j$, and phase logits are simply

$$\ell_j = A_j \mathbf{s}_j, \quad \ell_{j,p} = \sum_m A_{j,p,m} s_m.$$

An availability mask removes phases that cannot be chosen at the current decision. The remaining logits define a local categorical policy. This fixed sum is an explicit, inspectable interface and inductive bias; it is not claimed to be the only or optimal phase aggregation.

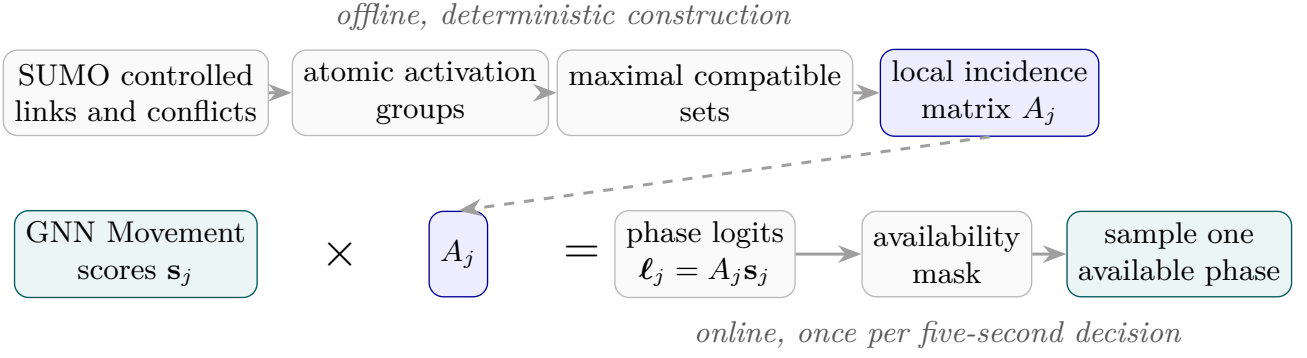


Figure 2: Offline phase construction is separate from online learning. The GNN supplies a variable-length Movement-score vector; a junction-local incidence matrix and temporary availability mask produce the action distribution.

3 Training and evaluation

PPO [7] is used as an optimizer and is not re-derived here. The anchored controller makes a decision every 5s, starts an immediate 3s yellow when switching, and requires one decision of minimum green. Training uses two message-passing blocks, four PPO epochs, 200 decisions per rollout, and entropy coefficient 0.001. Its local reward combines vehicle progress, discharge, braking, and gridlock terms with weights 1/10/10/0.02; global, throughput, flow, and direct switching terms are zero.

Mixed graph sizes require one practical correction: equal rollout counts are not equal PPO weight because larger graphs yield more junction actions per transition. Rollouts are therefore allocated to approximately balance the number of local action samples contributed by each training shape or city.

Three evaluations answer progressively less controlled questions:

- **Grid size and shape.** Three independently trained mixed-grid policies saw square and rectangular grids with longest axis at most five. They were evaluated without further learning on ten geometries, including 6×6 , using six fresh traffic seeds.
- **Signal coverage.** Removing controllers tests whether dense-signal training transfers to graphs where unsignalized pass-through edges replace controlled Movements. A separate 50%-coverage training run tests whether exposure to those gaps changes the outcome.
- **Cities.** One policy was trained with rollouts from Karlsruhe, Mannheim, Heidelberg, and Freiburg and evaluated on those scenarios plus Stuttgart. All runs use the fixed terminal checkpoint at iteration 60; training duration was not selected by stopping on an evaluation peak.

Sampled execution is primary because it matches the policy optimized by PPO. Greedy execution and full per-seed results are retained in the appendix. Max pressure is the main traffic-control baseline, with queue, fixed-time, and uniform-random policies as additional references.

4 Results

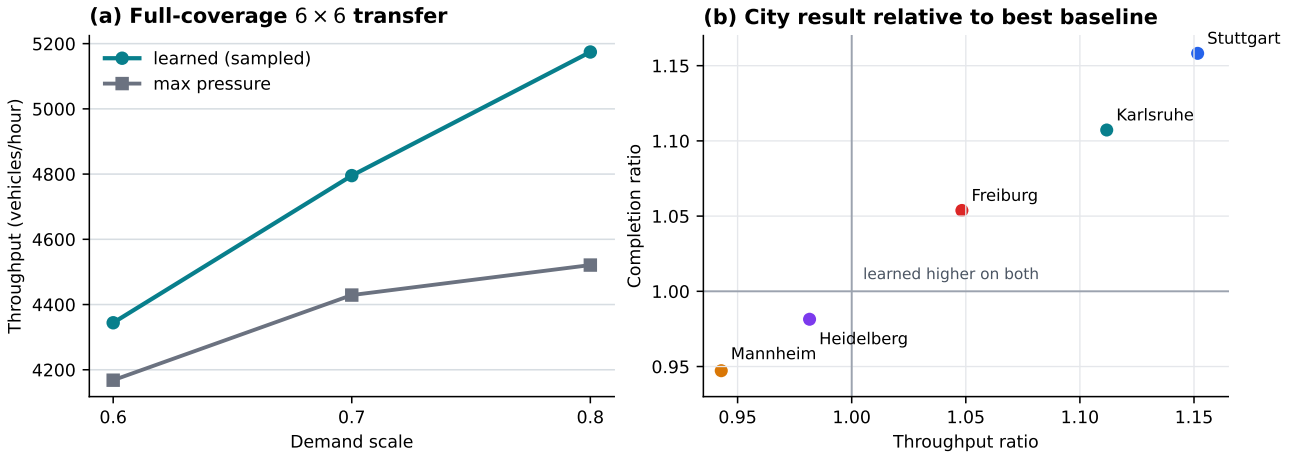


Figure 3: Succinct evaluation summary. Left: the sampled mixed-grid policy transfers to the larger full-coverage 6×6 grid across all evaluated demands (three training seeds and six fresh traffic seeds). Right: city throughput and completion relative to the strongest baseline for each metric, showing both successful and mixed cases.

Synthetic topology transfer. At full signal coverage, the learned policy retained useful behaviour as grid size and aspect ratio changed. On 6×6 , sampled throughput rose from 4344 to 5174 vehicles/hour across demand

scales 0.6 to 0.8, compared with 4168 to 4521 for max pressure. Completion improved by 3.5, 6.6, and 10.6 percentage points at the three demands. Transposed $5 \times 2/2 \times 5$ and $5 \times 3/3 \times 5$ pairs were also similar. These results support reuse across sizes and shapes generated by the same synthetic process; they do not establish transfer to arbitrary topology.

Coverage is distribution-sensitive. A policy trained only on dense signals deteriorated sharply when half or more eligible controllers were removed. Training on several 50%-coverage layouts recovered throughput and completion near max pressure on fresh 25% and 50% layouts, but did not yield uniformly good waiting density or higher-coverage behaviour. The graph can carry information through unsignalized gaps, but the policy must encounter the changed control distribution during training.

City behaviour is heterogeneous. The city graphs range from 286 to 708 LaneGroups, 271 to 508 Movements, and 41 to 84 policy-controlled junctions. One parameter set runs on all five. Sampled control is clearly positive in Karlsruhe and improves Stuttgart throughput/completion over every baseline. It is close to fixed time in Heidelberg, trails queue control in Mannheim, and improves Freiburg throughput and completion while not improving wait density. The five-city mean is 3345.6 vehicles/hour and 64.93% completion, versus 3036.8 and 59.37% for max pressure. The city experiment is one training run and should be read as a feasibility study, not a general city-transfer estimate.

5 Limitations and conclusion

Table 1: Intended interpretation of the evaluation evidence.

Evaluation	Supports	Does not establish
Synthetic grids	Size/shape reuse within the grid generator	Transfer to arbitrary topology
Signal coverage	Diagnosis of a control-distribution shift	Coverage-invariant control
Five cities	Execution feasibility and heterogeneous outcomes	A general city-transfer estimate

The architecture removes fixed graph and phase dimensions, but structural compatibility is not distributional robustness. Sparse signal coverage, different demand, and irregular city traffic can still change what the same local observations mean. Detector features are local, the phase sum assumes additive Movement utilities, and the conflict rules inherit the fidelity of the imported SUMO network. Sampled and greedy execution also differ materially, so deterministic argmax control cannot be substituted silently for the trained policy.

Within those boundaries, the implementation provides a compact way to connect a shared graph policy to heterogeneous traffic networks. LaneGroups give traffic state a directed corridor representation; Movements provide reusable controlled primitives; the second message-passing block introduces neighbouring Movement context; and deterministic phase construction keeps local compatibility outside the learned model. The synthetic experiments give the clearest evidence for size and shape reuse, while coverage and city results identify where training distribution remains decisive.

References

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A Implementation details

Table 2: Anchored training and control settings.

Setting	Value
Decision interval	5 s
Yellow transition	immediate, 3 s
Minimum green	one decision
Message-passing update blocks	2
Hidden dimension	64
Decisions per rollout	200
PPO epochs per update	4
Entropy coefficient	0.001
Progress/discharge/braking/gridlock	1/10/10/0.02
Global/flow/throughput/switch reward	0

Table 3: Structural diversity of the city scenarios. Phase range is the minimum–maximum number of selectable phases per policy-controlled junction.

City	Controllers	LaneGroups	Movements	Phase range
Karlsruhe	41	286	271	2–10
Mannheim	84	708	467	2–7
Stuttgart	69	519	508	2–12
Heidelberg	56	408	365	2–10
Freiburg	58	425	416	2–11

B Detailed evaluation

Table 4: Full-coverage 6×6 result. Learned values average three training seeds and six traffic seeds; max pressure averages the six traffic seeds.

Demand	Policy	Throughput/h	Completion	Wait density
0.6	learned sampled	4344	87.2%	0.077
	max pressure	4168	83.6%	0.082
0.7	learned sampled	4795	84.8%	0.114
	max pressure	4429	78.2%	0.119
0.8	learned sampled	5174	82.2%	0.175
	max pressure	4521	71.5%	0.131

Table 5: Iteration-60 city means over three traffic seeds. The baseline shown is the policy with the highest throughput in that city; wait density is reported in seconds per metre.

City	Policy	Throughput/h	Completion	Wait density
Karlsruhe	learned sampled	3452	93.86%	0.075
	max pressure	3105	84.77%	0.243
Mannheim	learned sampled	3200	53.78%	0.796
	queue	3394	56.78%	1.019
Stuttgart	learned sampled	4216	50.93%	0.907
	fixed time	3661	43.97%	0.741
Heidelberg	learned sampled	3190	77.69%	0.181
	fixed time	3250	79.16%	0.127
Freiburg	learned sampled	2670	48.39%	1.108
	fixed time	2547	45.92%	0.934

Sampled versus greedy. Greedy city control averaged 3182.6 vehicles/hour, 61.89% completion, and 0.980 s/m wait density, compared with 3345.6, 64.93%, and 0.613 s/m for sampled control. The gap motivates using the sampled policy as the primary result.

Coverage study. Dense-signal training transferred poorly to reduced coverage. A separate policy trained on five 50%-coverage 4×4 layouts matched max-pressure throughput and completion closely on fresh 25% and 50% layouts under sampled execution. At 75%, sampled throughput and completion were lower; greedy control retained throughput but accumulated more wait. This one-seed study diagnoses a training-distribution effect rather than establishing coverage-invariant control.

Evaluation units. Grid means should distinguish three independent policy-training seeds from the six traffic seeds used to evaluate each policy. The city study contains one policy-training seed and three traffic seeds. No claim in the main text relies on treating all traffic episodes as independent policy replications.